Business Insight Report
Student Performance Analysis

● Objective

The objective of this report is to analyze student performance data, identify important patterns related to academic scores, and provide practical recommendations to help improve students' academic outcomes.

● Dataset Summary

The dataset used for this analysis is the Student Performance dataset. It initially contained 25,000 records and 14 columns. After removing duplicate records, 15,000 records remained. The dataset includes information such as age, gender, parent education, study hours, attendance percentage, internet access, extra activities, study methods, Math scores, Science scores, English scores, overall scores, and final grades. These features help us explore patterns related to students' academic performance.

● Analysis Approach

The analysis was performed using Python in Google Colab. Pandas and NumPy were used to inspect, clean, and analyze the dataset. Duplicate records were removed, and missing values and data types were checked. Descriptive statistics were calculated to understand the data. Matplotlib and Seaborn were used to create visualizations, including a histogram, count plot, box plot, scatter plot, and correlation heatmap. These visualizations helped identify patterns and relationships between student-related factors and academic scores.

● Key Findings

1. **Study hours and overall scores:** Study hours have a strong positive correlation with overall scores (approximately 0.91). Students who study for more hours generally tend to have higher overall scores.
2. **Attendance and overall scores:** Attendance percentage has a weak positive correlation with overall scores (approximately 0.29). Attendance may be related to performance, but the relationship is relatively weak in this dataset.
3. **Subject scores:** Math, Science, and English scores have strong positive correlations with overall scores (approximately 0.88–0.89). Students' subject scores closely relate to their overall performance.

4. **Distribution of overall scores:** Most students' overall scores fall approximately between 40 and 90, while fewer students scored at the lower end. This shows that students have different performance levels, and some may need extra help with their studies.
5. **Gender distribution:** The dataset contains three gender categories: male, female, and other. Their counts are relatively balanced, with the other category having the highest count. The count plot shows the number of students in each category.
6. **Extra activities and performance:** The box plot comparing overall scores by extra-activity participation and gender shows no major differences between the groups, and the differences appear weak or unclear. This suggests that extra activity participation alone may not clearly distinguish students' overall scores in this dataset.

● Recommendations

1. Encourage students to follow regular study schedules and develop effective study habits.
2. Monitor attendance and provide appropriate support to students who frequently miss classes.
3. Review Math, Science, and English performance to identify students who may benefit from additional academic assistance.
4. Track students' scores regularly to identify changes in performance and provide timely support.
5. Collect additional information, such as assignment scores, previous academic results, and learning conditions, to support more detailed analysis.
6. Conduct further analysis before making decisions based on gender or participation in extra activities, because the current visualizations do not show clear performance differences between these groups.

● Limitations

This analysis has some limitations. The dataset does not include all factors that may affect student performance, such as teaching methods and students' previous academic results. Attendance shows only a weak positive relationship with overall scores, so its influence on performance cannot be clearly determined from this analysis. Similarly, the box plot shows no clear differences in performance based on extra-activity participation. The analysis identifies relationships between variables but cannot prove that one factor directly causes better scores. Therefore, additional data and further analysis are needed to understand student performance more fully.

Conclusion

The analysis identified a strong positive relationship between study hours and overall scores, while attendance showed a weaker positive relationship. Math, Science, and English scores were also strongly related to overall performance. These findings suggest that regular study habits and subject-level academic support are useful areas to investigate when planning student support. However, further analysis is needed to understand the causes of differences in student performance.